

OEC016 - Machine Learning and Predictive Analytics for Business. Introduction

Duration: 3 days

Description

This intensive three-day Machine Learning course for businesses provides a comprehensive, in-depth overview of the main Machine Learning techniques and methodologies, combining essential theory with extensive practice.

Designed for professionals with or without a technical background, the course covers the fundamentals of data preparation, exploratory analysis, predictive modelling and result interpretation. The blended approach combines brief theoretical introductions with extensive hands-on exercises guided using Orange, a widely used no-code Machine Learning tool renowned for its exceptional ease of use and requiring no programming knowledge.

Participants can apply the concepts directly by building Machine Learning workflows and analysing real-world business datasets.

The course aims to provide immediately applicable skills for optimising business processes, improving strategic decisions and fully exploiting the potential of data through Machine Learning. Using Orange as a no-code tool reduces the technical complexity of programming, allowing participants to focus on analysing and interpreting results.

By the end of the course, participants will be able to apply Machine Learning methodologies in their own business contexts, using practical tools to make data-driven decisions.

Target audience

The course is intended for professionals, analysts, data scientists, software engineers and managers who wish to explore the use of Machine Learning for data analysis and business trend forecasting. It is suitable both for participants with a technical background and for anyone interested in understanding how to implement ML solutions in business processes.

Prerequisites

A logical and mathematical inclination is certainly useful for participants in this course, although it is not strictly essential.

Contents

Module 1: Introduction to Machine Learning (ML)

- Definition and overview of ML
- A brief history of ML
- Differences between supervised and unsupervised ML
- ML vs AI vs Data Mining vs Statistics: how not to confuse them
- Familiar ML examples. The leading role of Google, Amazon and Facebook
- Why ML can help companies make better decisions
- Developing a data-driven mindset, supported by intuition (synergy)
- Which areas of a company's business are suitable for (predictive) ML?
- Key ML terminology

Module 2: Data Loading and Pre-processing

- Standardisation and normalisation
- Detecting and handling missing values
- Identifying and handling outliers

Module 3: Exploratory Data Analysis (EDA)

- Variable types: numerical and categorical
- Datasets (in various formats)
- Numerical correlation and correlation matrices
- Dependence and independence of categorical variables (chi-square test)
- Charts: scatter plots, box plots, histograms, bar charts, line charts, heatmaps
- Interactive visualisations

Module 4: Predictive Modelling

- Data fitting
- The decision boundary
- Regression models: linear and logistic
- Classification models: decision trees, random forests, KNN, SVM, Naive Bayes
- Neural networks
- Simple vs complex models
- Rigid vs flexible models
- Imbalanced classes: what to do

Module 5: Model Evaluation

- Training, testing and cross-validation
- Metrics: accuracy, confusion matrix, R^2 , RMSE
- Overfitting and omitted variables (the two main model problems): how to avoid them
- The important bias-variance trade-off

Module 6: Feature Selection

- Manual and automatic techniques

Module 7: Deployment of ML Models

- Deployment tools: APIs, containerisation (Docker)
- Integrating models into business systems
- MLOps
- Monitoring and maintaining models in production

Module 8: Unsupervised Machine Learning

- Clustering: K-means, hierarchical, DBSCAN, t-SNE
- Distances and distance matrices
- PCA (Principal Component Analysis)

Module 9: Time Series

- Trend and pattern analysis
- Forecasting future values (regression, exponential smoothing, ARIMA)

Module 10: Hands-on Exercises with Orange

- Introduction to Orange
- Creating workflows in Orange
- Analysis of business cases

Module 11: Exporting Results

- Exports, reports and documentation

Module 12: ML Technologies and Tools

- Open source: Python, R, scikit-learn, Google's TensorFlow, PyTorch, Weka, Orange
- Proprietary: Google, Microsoft
- No-code and low-code tools
- Python, R and Excel
- Microsoft

Module 13: Getting Started Instructions

Module 14: Bibliography and Web Resources